SEMIPRIME RIGIDITY AND GROWTH ALONG POWERS IN INTEGER-SIDED CYCLIC POLYGONS

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ABSTRACT. For a positive integer n, let $a(n)$ be the largest number of sides of a simple cyclic polygon of circumradius n whose side lengths are positive integers. For primes p, q we show that $a(pq) \in \{6, 8, 18\}$ and characterize each value by conditions on Gaussian and Eisenstein integers. For every integer $N \geq 2$, on the other hand, $a(N^e) \to \infty$ and $\limsup_{e\to\infty} a(N^e)/e \geq 2$; a quadratic chord tower combined with Farey neighbors gives explicit polygons, for instance $a(8192) \geq 38$. We also show that an odd-sided polygon of radius n exists if and only if n has a prime divisor congruent to 1 (mod 4), and that $a(n)$ is even whenever at most one distinct prime divisor of n is congruent to 1 (mod 4). The proofs use a splitting lemma for angle relations by squarefree class, arithmetic in imaginary quadratic fields, bounded zero-sum problems, and a one-dimensional Farey argument. No computer-assisted step is used.

1. INTRODUCTION

Fix a positive integer n and consider simple polygons inscribed in a circle of radius n all of whose sides have positive integer length. How many sides can such a polygon have? A regular hexagon, with side n, always gives six. At radius 65 there is an octagon with sides $32, 50, 50, 66, 32, 50, 50, 66$ (Example 7.2); at radius 91 there is an eighteen-gon whose sides $7, 26, 61$ each occur six times (Example 7.1); and at radius $2^{13} = 8192$ the construction of Section 5 gives a polygon with 38 sides.

A chord of length s in a circle of radius n subtends a central angle whose half is

$$\alpha_n(s) = \arcsin\left(\frac{s}{2n}\right) \in (0, \pi/2] \qquad (1 \leq s \leq 2n), \tag{1}$$

and a polygon closes up precisely when such angles satisfy one exact additive relation (Lemma 2.1). Which angles occur, and which additive relations hold among them, depends on the arithmetic of n. Much of the classical literature on cyclic polygons concerns areas and rationality conditions [6, 4, 1], and trigonometric Diophantine relations have a separate tradition [2, 3]. Here we study the *maximum number of integer chords that close up* at a prescribed integer circumradius.

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Definition 1.1. Let $a(n)$ be the greatest integer $k \geq 3$ for which there is a simple cyclic polygon with k distinct vertices, circumradius n, and positive integer side lengths.

The regular hexagon has integer side length n, so $a(n) \geq 6$. The maximum exists: every side subtends a central angle of at least $2\alpha_n(1)$, so the full turn bounds the number of sides. The sequence is OEIS A398490; its associated counting triangle is A398489 [5].

Results. Semiprime radii. Our first result determines $a(n)$ when n is a product of two primes, not necessarily distinct. The exceptional values are governed by angles arising from the Eisenstein and Gaussian integers. To state the criterion, let $r \equiv 1 \pmod 3$ be prime and let (s_r, u_r) be the solution of

$$s_r^2 + 3u_r^2 = 4r^2, \qquad 0 < s_r < r, \quad u_r > 0; \tag{2}$$

existence and uniqueness are proved in Lemma 4.4. Write

$$x_r = \arcsin\left(\frac{s_r}{2r}\right) \in (0, \pi/6). \tag{3}$$

Theorem A (Semiprime radii)**.** *Let $p \leq q$ be primes and consider the conditions*

(E) $p \equiv q \equiv 1 \pmod 3$ (so p, q are odd) and $x_p + x_q < \pi/6$;
(G) $p < q$ and $p \equiv q \equiv 1 \pmod 4$.

Then

$$a(pq) = \begin{cases} 18, & \text{if (E) holds,} \\ 8, & \text{if (G) holds but (E) does not,} \\ 6, & \text{otherwise.} \end{cases}$$

Moreover, $x_p + x_q < \pi/6$ is equivalent to the integer inequality

$$3(u_p s_q + u_q s_p) < 3u_p u_q - s_p s_q. \tag{4}$$

In particular,

$$a(pq) \in \{6, 8, 18\}, \qquad a(2p) = a(3p) = 6$$

for every prime p.

For a semiprime radius, each integer chord determines a vector of prime-ideal valuations in a small box, and an exact closing relation becomes a zero-sum relation among these vectors. In an imaginary quadratic field with only the units ± 1 this does not allow more than six sides; the additional roots of unity in the Gaussian and Eisenstein fields allow eight and eighteen sides. For example, $a(49) = 18$, $a(65) = 8$, $a(91) = 18$, while $a(361) = 6$ (Section 7). For $n = 1$ and every prime n one has $a(n) = 6$ (Remark 4.9). Semiprime radii are therefore the first level, classified by the number of prime factors counted with multiplicity, at which the maximum can exceed six.

Powers of a fixed integer. Restricting the prime divisors of the radius does not bound $a(n)$: successive powers of a fixed integer admit polygons with arbitrarily many sides.

Theorem B (Growth along powers)**.** *For every integer $N \geq 2$,*

$$a(N^e) \longrightarrow \infty \qquad (e \to \infty),$$

and more quantitatively

$$\limsup_{e\to\infty} \frac{a(N^e)}{e} \geq 2. \tag{5}$$

The proof builds integer chords from powers of one imaginary quadratic integer; neighboring fractions on a Stern–Brocot path give the exact cancellation of angles that closes the polygon. For $N = 2$ it yields a 38-gon at radius $8192 = 2^{13}$ (Example 5.3).

Parity. Theorem 3.6 shows that an integer-sided cyclic polygon with an odd number of sides exists at radius n if and only if n has a prime divisor congruent to 1 modulo 4. Whether an *extremal* polygon can have an odd number of sides is a subtler question. Define

$$\omega_1(n) = \#\{q : q \mid n,\ q \text{ prime},\ q \equiv 1 \pmod 4\},$$

where distinct primes are counted only once.

Theorem C (Parity in Gaussian rank one)**.** *If $\omega_1(n) \leq 1$, then $a(n)$ is even. In particular,*

$$a(p^e) \equiv 0 \pmod 2$$

for every prime p and every $e \geq 1$.

Organization. All three theorems rest on exact additive relations among the angles $\alpha_n(s)$. Section 2 gives the closing criterion. Section 3 shows that such relations split by squarefree class, which also yields the odd-sided criterion. Section 4 proves Theorem A: zero-sum relations bound the number of chords per unit angle in each class, and explicit integer constructions attain the bounds in the Gaussian and Eisenstein classes. Sections 5 and 6 prove Theorems B and C. Section 7 contains examples and Section 8 open questions.

2. The geometric criterion

Lemma 2.1 (Minor- and major-arc criterion)**.** *A multiset of $k \geq 3$ positive integers $\{s_1, \ldots, s_k\}$, with $s_i \leq 2n$, is realizable as the side-length multiset of a simple cyclic polygon of radius n if and only if either*

$$\sum_{i=1}^{k} \alpha_n(s_i) = \pi, \tag{6}$$

or

$$\alpha_n(s_j) = \sum_{i\neq j} \alpha_n(s_i) \tag{7}$$

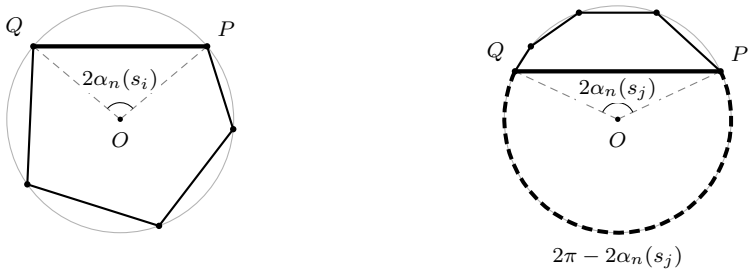
for some j.

Proof. For points on a circle, the boundary edges of a simple polygon follow their circular order, up to reversal. Indeed, if an edge joined two nonconsecutive vertices, the remaining vertices on the two open arcs between its endpoints would lie on opposite sides of that chord. The remaining boundary path, which avoids the chord's endpoints, would have to connect the two

sides and therefore cross the chord; this contradicts simplicity. Thus the successive boundary arcs form an ordered partition of the full circle and their central angles sum to 2π. If every such arc has length at most π, its central angle is $2\alpha_n(s_i)$, giving (6). Otherwise exactly one arc has length greater than π (two such arcs would already exceed 2π), and its central angle is $2\pi - 2\alpha_n(s_j)$; the other arcs are minor, and the central angles sum to 2π exactly when (7) holds.

Conversely, suppose $k \geq 3$ and one of these equations holds. Arrange on the circle consecutive arcs with central angles $2\alpha_n(s_i)$ in the first case, and $2\alpha_n(s_i)$ $(i \neq j)$ together with $2\pi - 2\alpha_n(s_j) > 0$ in the second. Each angle is strictly positive and their sum is 2π; therefore the cumulative endpoints give k distinct circle vertices in cyclic order. Joining consecutive vertices yields a convex, hence simple, cyclic k-gon, and the chord subtending an arc of central angle $\theta \in (0, 2\pi)$ has length $2n\sin(\theta/2)$, which equals the prescribed s_i in both cases. □

Remark 2.2. A side of length $2n$ (a *diameter*) has $\alpha_n(2n) = \pi/2$. At most one side can be a diameter, because the other $k - 1 \geq 2$ half-angles are positive. A relation (7) with $\alpha_n(s_j) = \pi/2$ is the same as (6). Hence every polygon is of exactly one of three kinds: a polygon *with a diameter*, a *minor-arc polygon without diameter* (relation (6)), or a *major-arc polygon* (relation (7) with $\alpha_n(s_j) < \pi/2$). Figure 1 illustrates the last two kinds.



(a) All boundary arcs are minor. (b) One boundary arc is major.

FIGURE 1. Minor and major configurations in Lemma 2.1. The heavy chord PQ has length s_i in (a) and s_j in (b); dashed segments join P and Q to the center O. In (b) the boundary arc from Q to P (dashed circular arc) has central angle $2\pi - 2\alpha_n(s_j)$, while the other sides together subtend the central angle $2\alpha_n(s_j)$ of PQ. The diagram was drafted with AI-assisted TikZ code from this geometric construction.

Definition 2.3 (Blocks)**.** Put $H = \pi/2$. A *block* at radius n is a finite nonempty multiset of half-angles $\alpha_n(s)$ of *non-diameter* sides $(1 \leq s < 2n)$; its *total* is the sum of its members. A block is a *π-block* or an *H-block* if its total is π or H, respectively, and a *rational block* if its total lies in $\pi\mathbb{Q}$. A block is *pure* in a squarefree class d (defined in Section 3) if all its members lie in class d.

Thus (6) says that the sides of a minor-arc polygon without diameter form a π-block, and a polygon with a diameter consists of the diameter together with an H-block (Remark 2.2); relation (7) is the *major-arc relation*.

We shall also use the strict monotonicity

$$\frac{d}{ds}\alpha_n(s) = \frac{1}{\sqrt{4n^2 - s^2}} > 0. \tag{8}$$

Thus a half-angle uniquely determines its side length at fixed radius.

Proposition 2.4 (Scaling)**.** *For all positive integers m, n,*

$$a(mn) \geq a(n).$$

Proof. Scale every side length of a polygon of radius n by m. All central angles are unchanged, and the new polygon has radius mn. □

3. Squarefree classes and elementary splitting

We group chords by the squarefree part of the complementary chord equation and show that rational angle relations split across these classes. This is the arithmetic input for Theorems A and C.

For a non-diameter side s at radius n, write

$$4n^2 - s^2 = du^2, \tag{9}$$

where $d > 0$ is squarefree and u is positive; we say that s (or its half-angle) lies in the *class* d. Recall $H = \pi/2$ and set $\beta_n(s) = H - \alpha_n(s) \in (0, H)$. Then

$$\tan \beta_n(s) = \frac{u}{s}\sqrt{d}. \tag{10}$$

In particular, $\tan^2 \beta_n(s) = du^2/s^2$ is rational, and the tangent belongs to a fixed squarefree class d. Angles with rational squared tangent are the subject of Conway, Radin, and Sadun [3]; we prove the splitting property needed below directly, using elementary ideal theory and Galois characters. The classes $d = 1$ (*Gaussian*) and $d = 3$ (*Eisenstein*) are special because $\mathbb{Q}(\sqrt{-d})$ then has more than two roots of unity; all other classes are called *ordinary*.

For later normalization, put

$$\kappa = \gcd(s, u), \qquad A = \frac{s}{\kappa}, \qquad B = \frac{u}{\kappa}, \qquad M = \frac{2n}{\kappa}. \tag{11}$$

Because κ^2 divides $s^2 + du^2 = (2n)^2$, one has $\kappa \mid 2n$; thus M is a positive integer and $M \mid 2n$. Moreover,

$$\gcd(A, B) = 1, \qquad A^2 + dB^2 = M^2. \tag{12}$$

3.1. **Ideal components.** With the normalization (11) and (12), put $K = \mathbb{Q}(\sqrt{-d})$, $z = A + B\sqrt{-d}$, and $w = z/M$. Then $|w| = 1$, $\arg w = \beta_n(s) = H - \alpha_n(s) \in (0, H)$, and

$$(z)(\bar{z}) = (M)^2. \tag{13}$$

For each rational prime r that splits as $(r) = \mathfrak{r}\bar{\mathfrak{r}}$, fix once and for all an ordering of the two prime ideals and put

$$v_r(w) = \tfrac{1}{2}(\mathrm{ord}_{\mathfrak{r}}(w) - \mathrm{ord}_{\bar{\mathfrak{r}}}(w)) = \tfrac{1}{2}(\mathrm{ord}_{\mathfrak{r}}(z) - \mathrm{ord}_{\bar{\mathfrak{r}}}(z)).$$

The equality (13) makes the two exponents of z sum to $2\operatorname{ord}_r M$; hence their difference is even and $v_r(w) \in \mathbb{Z}$. The fractional ideal (w) therefore has exponents $v_r(w), -v_r(w)$ at $\mathfrak{r}, \bar{\mathfrak{r}}$. At inert and ramified primes the exponents of (w) vanish, and so do those at primes not dividing M. In particular, only split primes dividing $2n$ may contribute. We call $v_r(w)$ the *r-component* of the chord. The component vectors of different chords are always computed with the same fixed orientation of each split prime-ideal pair.

Lemma 3.1 (Component criterion)**.** *Let $w_1, \dots, w_m$ be the numbers attached as above to non-diameter sides (possibly at different radii) of one* fixed class *d, with half-angles α_i, and let $\varepsilon_i \in \mathbb{Z}$. Then*

$$\sum_i \varepsilon_i \alpha_i \in \pi\mathbb{Q} \quad \Longleftrightarrow \quad \sum_i \varepsilon_i v_r(w_i) = 0 \text{ for every split prime } r. \tag{14}$$

Moreover, in that case $\prod_i w_i^{\varepsilon_i}$ is a root of unity of K. The unit group of $\mathcal{O}_K$ is $\{\pm 1\}$ for $d \neq 1, 3$, μ_4 for $d = 1$, and μ_6 for $d = 3$.

Proof. The product $W = \prod_i w_i^{\varepsilon_i} \in K^\times$ has absolute value 1 and argument $\sum_i \varepsilon_i \beta_i = (\sum_i \varepsilon_i) H - \sum_i \varepsilon_i \alpha_i$. Since $H \in \pi\mathbb{Q}$, the left-hand condition in (14) says exactly that $\arg W \in \pi\mathbb{Q}$, i.e. that W is a root of unity. The fractional ideal of W is $\prod_r (\mathfrak{r}/\bar{\mathfrak{r}})^{\sum_i \varepsilon_i v_r(w_i)}$ by the preceding paragraph, and this vanishes if and only if all exponents vanish. If W is a root of unity its ideal is trivial, which gives the forward implication. Conversely, if the ideal of W is trivial, then W is a unit of $\mathcal{O}_K$, hence a root of unity because the unit group is finite. □

Lemma 3.2 (Elementary splitting of squarefree classes)**.** *Let α_i be finitely many non-diameter integer-chord half-angles, possibly at different radii, and let $\varepsilon_i \in \mathbb{Z}$. If $\sum_i \varepsilon_i \alpha_i \in \pi\mathbb{Q}$, then for every squarefree d the restricted sum $\sum_{i:d_i=d} \varepsilon_i \alpha_i$ belongs to $\pi\mathbb{Q}$.*

Proof. As $\beta_i = H - \alpha_i$, the hypothesis also gives $\sum_i \varepsilon_i \beta_i \in \pi\mathbb{Q}$. For every occurring squarefree d, put $K_d = \mathbb{Q}(\sqrt{-d})$ and $W_d = \prod_{i:d_i=d} w_i^{\varepsilon_i} \in K_d$. We have $|W_d| = 1$ and $\prod_d W_d = \zeta$, a root of unity. Let L be the compositum of the fields K_d and $G = \operatorname{Gal}(L/\mathbb{Q})$. It is a finite elementary abelian 2-group. Each field K_d defines a character $\chi_d : G \to \{\pm 1\}$: a Galois automorphism acts on K_d as the identity when $\chi_d = 1$ and as complex conjugation when $\chi_d = -1$. Distinct squarefree d give distinct characters, since $\ker \chi_d = \operatorname{Gal}(L/K_d)$ determines K_d. Since $\overline{W_d} = W_d^{-1}$, $\sigma(W_d) = W_d^{\chi_d(\sigma)}$. Fix a class d_0. Apply every $\sigma \in G$ to $\prod_d W_d = \zeta$, take the $\chi_{d_0}(\sigma)$-th power, and multiply over σ; this yields

$$\prod_{\sigma \in G} \sigma(\zeta)^{\chi_{d_0}(\sigma)} = \prod_d W_d^{\sum_{\sigma \in G} \chi_{d_0}(\sigma)\chi_d(\sigma)} = W_{d_0}^{|G|},$$

by the orthogonality of distinct characters of the finite abelian group G. The left side is a root of unity; hence W_{d_0} is a root of unity. Thus $\sum_{i:d_i=d_0} \varepsilon_i \beta_i \in \pi\mathbb{Q}$, and subtracting from the corresponding integral multiple of H proves the assertion for the α_i. □

The next lemma locates the rational sub-sums of Lemma 3.2. It is used for the parity principle, for major-arc relations, and in Section 6.

Lemma 3.3 (Unit lattices of rational sub-sums)**.** *Let α_i and ε_i be as in Lemma 3.2, with $\sum_i \varepsilon_i\alpha_i \in \pi\mathbb{Q}$. For a squarefree class d put*

$$\mathcal{A}_d = \sum_{i:\, d_i=d} \varepsilon_i\alpha_i, \qquad m_d = \sum_{i:\, d_i=d} \varepsilon_i, \qquad U = \frac{\pi}{6}.$$

Then $\mathcal{A}_d \in \pi\mathbb{Q}$ and:

(i) if $d \neq 1, 3$, then $\mathcal{A}_d \in H\mathbb{Z}$ and $\mathcal{A}_d/H \equiv m_d \pmod 2$;
(ii) if $d = 1$, then $\mathcal{A}_d \in H\mathbb{Z}$;
(iii) if $d = 3$, then $\mathcal{A}_d \in U\mathbb{Z}$ and $\mathcal{A}_d/U \equiv m_d \pmod 2$.

Proof. By Lemma 3.2, $\mathcal{A}_d \in \pi\mathbb{Q}$, so $\mathcal{B}_d = \sum_{i:d_i=d} \varepsilon_i\beta_i = m_dH - \mathcal{A}_d \in \pi\mathbb{Q}$, and by Lemma 3.1 the number $W_d = \prod_{d_i=d} w_i^{\varepsilon_i}$ is a root of unity of $\mathbb{Q}(\sqrt{-d})$ with argument $\mathcal{B}_d$ (modulo 2π). Hence $\mathcal{B}_d \in \pi\mathbb{Z} = 2H\mathbb{Z}$ for $d \neq 1, 3$, $\mathcal{B}_d \in H\mathbb{Z}$ for $d = 1$, and $\mathcal{B}_d \in (\pi/3)\mathbb{Z} = 2U\mathbb{Z}$ for $d = 3$. Now use $\mathcal{A}_d = m_dH - \mathcal{B}_d$, and for $d = 3$ use $H = 3U$ and $3m_d \equiv m_d \pmod 2$. □

The following parity principle gives the odd-sided criterion and isolates the Gaussian class as the only obstruction to evenness.

Lemma 3.4 (Non-Gaussian parity principle)**.** *Let $\alpha_1, \ldots, \alpha_m$ be half-angles of non-diameter integer chords of the same radius, none belonging to the Gaussian class $d = 1$. Let $\varepsilon_i \in \{1, -1\}$, and suppose*

$$\sum_{i=1}^{m} \varepsilon_i\alpha_i = tH, \qquad H = \frac{\pi}{2}, \qquad t \in \mathbb{Z}.$$

Then

$$\sum_{i=1}^{m} \varepsilon_i \equiv t \pmod 2.$$

Proof. Lemma 3.3 applies with the sums $\mathcal{A}_d$, $d \neq 1$. For $d \neq 1, 3$ we have $\mathcal{A}_d/H \equiv m_d \pmod 2$. For $d = 3$ we have only $\mathcal{A}_3 \in U\mathbb{Z}$; but $\mathcal{A}_3 = tH - \sum_{d\neq 3} \mathcal{A}_d \in H\mathbb{Z}$, so $\mathcal{A}_3 = \ell H = 3\ell U$ for an integer ℓ, and $\ell \equiv 3\ell = \mathcal{A}_3/U \equiv m_3 \pmod 2$. Thus $\mathcal{A}_d/H \equiv m_d \pmod 2$ for every occurring class d. Summing over d gives $t = \sum_d \mathcal{A}_d/H \equiv \sum_d m_d = \sum_i \varepsilon_i \pmod 2$. □

Proposition 3.5 (Parity without Gaussian chords)**.** *At any radius n, a simple integer-sided cyclic polygon containing no non-diameter side of squarefree class $d = 1$ has an even number of sides.*

Proof. Let k be the number of sides. By Remark 2.2 there are three kinds of polygon. *Minor-arc polygon without diameter:* Lemma 3.4 with all signs $+1$ and $t = 2$ gives $k \equiv 2 \equiv 0 \pmod 2$. *Major-arc polygon:* give the target α_j the sign $+1$ and each other side the sign -1; the signed sum is 0, so $t = 0$ and $1 - (k - 1) = 2 - k \equiv 0$, i.e. k is even. *Polygon with a diameter:* the remaining $k - 1$ non-diameter half-angles sum to H, so Lemma 3.4 with all signs $+1$ and $t = 1$ shows that $k - 1$ is odd, hence k is even. □

Theorem 3.6 (Odd-sided polygons)**.** *For a positive integer n, the following are equivalent:*

(i) there exists an odd-sided integer-sided cyclic polygon of radius n;
(ii) there exists such a triangle;
(iii) n has a prime divisor congruent to $1 \pmod 4$.

Proof. (ii)⇒(i) is immediate.

(iii)⇒(ii). Let $q \equiv 1 \pmod 4$ be a prime divisor of n, say $n = qm$. By Fermat's theorem $q = a^2 + b^2$ with positive integers $a \neq b$ (q is odd). Then $2m(a^2 - b^2)$ (in absolute value), $4mab$ and $2mq = 2n$ are the sides of a nondegenerate right triangle with integer sides, since $(a^2-b^2)^2 + (2ab)^2 = q^2$. Its circumradius is half its hypotenuse, namely n.

(i)⇒(iii), contrapositively. Suppose that n has no prime divisor congruent to 1 (mod 4). Then there is no non-diameter Gaussian side. Indeed, such a side would give $s^2 + u^2 = (2n)^2$ with $s, u > 0$; dividing by $\gcd(s,u)^2$ gives a primitive Pythagorean triple whose hypotenuse $2n/\gcd(s,u) > 1$ is odd and has all its prime divisors congruent to 1 (mod 4), and these divide n. Hence Proposition 3.5 shows that every admissible polygon has an even number of sides, so no odd-sided polygon exists. □

4. Semiprime radii

The regular hexagon gives $a(n) \geq 6$. For semiprime n we bound the number of chords per unit angle within each squarefree class. Zero-sum relations among component vectors (§4.1) give the bound T/U for a block of total T in every ordinary class (§4.2), where $U = \pi/6$. In the Gaussian and Eisenstein classes (§§4.3–4.4) the extra units allow more chords per unit angle: we list all candidate half-angles as integral combinations of one unit angle and the angles attached to the split primes, give each candidate an integer weight $\Phi \geq 1$ that is additive over rational blocks, and obtain the densities 4/3 and 3. Explicit integer polygons attain these densities. Class splitting adds the bounds over all classes (§4.5, where major arcs and polygons with a diameter are also treated), and §4.6 assembles the proof of Theorem A.

Set $H = \pi/2$ and $U = \pi/6$, so $H = 3U$. Blocks, rational blocks and pure blocks are as in Definition 2.3; the total of a rational block is always positive.

4.1. **Component bounds and zero-sum atoms.**

Lemma 4.1 (Uniform component bounds)**.** *For semiprime n, the following bounds hold for each non-diameter chord in a fixed squarefree class:*

- *(i) $n = pq$, $p < q$ odd primes: $(v_p, v_q) \in \{-1,0,1\}^2$, with no nonzero component at 2;*
- *(ii) $n = p^2$, p odd: $v_p \in \{-2,-1,0,1,2\}$, with no nonzero component at 2;*
- *(iii) $n = 2p$, p odd: $(v_2, v_p) \in \{-1,0,1\}^2$, with a coordinate that can be nonzero only when that prime splits;*
- *(iv) $n = 4$: only the prime 2 can contribute, with $|v_2| \leq 2$;*
- *(v) $n = p$ prime (used only in Remark 4.9): $v_p \in \{-1,0,1\}$ and no other component occurs.*

Proof. Let s be a non-diameter side in class d, with κ, A, B, M as in (11)–(12), $K = \mathbb{Q}(\sqrt{-d})$ and $z = A + B\sqrt{-d}$. Let r be a rational prime that splits, $(r) = \mathfrak{r}\bar{\mathfrak{r}}$, and put $e_1 = \operatorname{ord}_{\mathfrak{r}} z$, $e_2 = \operatorname{ord}_{\bar{\mathfrak{r}}} z$. By (13), $e_1 + e_2 = 2\operatorname{ord}_r M$ and $v_r = (e_1 - e_2)/2$; in particular $|v_r| \leq \operatorname{ord}_r M$, and $v_r = 0$ if $r \nmid M$.

Odd split primes. Let r be odd with $r \mid M$. If $e_1, e_2 \geq 1$, then $z \in \mathfrak{r} \cap \bar{\mathfrak{r}} = r\mathcal{O}_K$. Since $2\mathcal{O}_K \subseteq \mathbb{Z}[\sqrt{-d}]$, we get $2z \in r\,\mathbb{Z}[\sqrt{-d}]$, i.e. $r \mid 2A$ and $r \mid 2B$,

hence $r \mid A, B$ because r is odd; this contradicts $\gcd(A, B) = 1$. Therefore $\min(e_1, e_2) = 0$ and $|v_r| = \operatorname{ord}_r M \le \operatorname{ord}_r(2n)$.

The prime 2. The prime 2 splits in K if and only if $d \equiv 7 \pmod 8$. Suppose it splits and $2 \mid M$. Since d is odd, $A^2 + dB^2 = M^2$ is even and $\gcd(A, B) = 1$, both A, B are odd; then $A^2 + dB^2 \equiv 1 + 7 \equiv 0 \pmod 8$, so $4 \mid M$. With $\eta = (1 + \sqrt{-d})/2 \in \mathcal{O}_K$ we have $z = (A - B) + 2B\eta \in 2\mathcal{O}_K$, so $e_1, e_2 \ge 1$.

Conclusion. Recall $M \mid 2n$. In case (i), $\operatorname{ord}_p M, \operatorname{ord}_q M \le 1$, and $\operatorname{ord}_2 M \le 1$ excludes a component at 2 (it would need $4 \mid M$). In case (ii), $\operatorname{ord}_p M \le 2$ and again 2 cannot contribute. In case (iii), $|v_p| \le 1$; if 2 splits and $2 \mid M$, then $4 \mid M \mid 4p$ forces $\operatorname{ord}_2 M = 2$, so $e_1 + e_2 = 4$ with $e_1, e_2 \ge 1$ and $|v_2| \le 1$. In case (iv) only 2 divides $2n = 8$; if it splits and $2 \mid M$, then $\operatorname{ord}_2 M \in \{2, 3\}$, $e_1 + e_2 \in \{4, 6\}$ with $e_1, e_2 \ge 1$, and hence $|v_2| \le 1$ resp. ≤ 2. In case (v), $|v_p| \le 1$ for odd p by the first argument and, as $\operatorname{ord}_2(2p) = 1$, no component at 2 occurs; for $p = 2$ we have $M \mid 4$, and a component at 2 forces $4 \mid M$, $e_1 + e_2 = 4$ and $|v_2| \le 1$. □

Lemma 4.2 (Zero-sum atom lengths)**.** *A minimal nonempty zero-sum multiset of nonzero vectors from* $\{-1, 0, 1\}^2$ *has at most four members. The bound is sharp. A minimal zero sum of nonzero vectors in* $\{-2, -1, 1, 2\}$ *has at most three; in* $\{-1, 1\}$ *it has exactly two.*

Proof. If an opposite pair occurs, minimality leaves precisely that pair. Otherwise the support contains at most one of each of the four pairs of opposite directions, so it has at most four directions. Suppose four directions occur. After an integral isometry of the square we may take the two diagonal directions to be $(1, 1)$ and $(1, -1)$, with respective positive multiplicities c, d. Their total is $(c + d, c - d)$. The horizontal direction must then be $(-1, 0)$ with multiplicity $c + d$; the vertical direction has multiplicity $|c - d|$. Since all four directions occur, $c \ne d$. If $c > d$, the multiset contains the zero-sum triple $\{(1, 1), (-1, 0), (0, -1)\}$; if $d > c$, it contains $\{(1, -1), (-1, 0), (0, 1)\}$. Either is a proper nonempty zero-sum submultiset, a contradiction. Thus the support has at most three directions.

With no opposite pair, a positive dependence cannot have just one or two distinct directions. For three noncollinear directions u, v, w, the unique primitive positive dependence has coefficients the absolute determinants $|\det(v, w)|$, $|\det(w, u)|$, $|\det(u, v)|$, divided by their gcd. Each determinant is at most 2. Its absolute value can equal 2 only for a pair of diagonal vectors. With no opposite pair, there can be at most two diagonal directions in the support, so at most one coefficient can be 2. Consequently the coefficients sum to at most $2 + 1 + 1 = 4$. The multiset $\{(1, 0), (1, 0), (-1, 1), (-1, -1)\}$ shows sharpness. In one dimension the minimal possibilities, up to a global sign, are $\{1, -1\}$, $\{2, -2\}$, $\{2, -1, -1\}$ (and $\{-2, 1, 1\}$). □

4.2. **Ordinary classes: a universal density bound.** For $d \ne 1, 3$, the only units of $\mathbb{Q}(\sqrt{-d})$ are ± 1. This converts the bound on zero-sum atom lengths into a uniform bound on the number of chords per unit of angular measure.

Lemma 4.3 (No ordinary class exceeds the hexagonal density)**.** *For any semiprime radius* n *and any rational block of* k *half-angles, pure in one*

squarefree class $d \neq 1, 3$ and with total $T \in \pi\mathbb{Q}$, one has

$$k \leq T/U. \tag{15}$$

In particular, every such block has $T \geq H$.

Proof. By (14) the component vectors of the members sum to zero. Decompose them into minimal zero-sum atoms (repeatedly split off a minimal nonempty zero-sum submultiset), allowing zero vectors as singleton atoms. Let $w_1, \ldots, w_m$ be the numbers w of the members of one atom. By Lemma 3.1 their product is a root of unity of $\mathbb{Q}(\sqrt{-d})$, hence ± 1, so $\sum_i \beta_i \in \pi\mathbb{Z}$, where $0 < \beta_i < H$. A singleton atom ($m = 1$) would give $\beta_1 \in \pi\mathbb{Z}$, which is impossible. For $m = 2$ we would get $\beta_1 + \beta_2 \in \pi\mathbb{Z} \cap (0, \pi) = \emptyset$. By Lemmas 4.1 and 4.2 every remaining atom has $m = 3$ or $m = 4$ members (for $n = p^2$ or $n = 4$ only $m = 3$ occurs, and for prime n no atom remains). Its β-sum is a multiple of π in $(0, mH)$, hence exactly π, and its half-angle sum is therefore $mH - \pi = (m-2)H$. Since $m \leq 3(m-2)$ for $m = 3, 4$, this atom satisfies $m \leq (m-2)H/U$. Summing over the atoms gives (15); each atom contributes at least H to T. □

4.3. **Gaussian classes and integral constructions.**
Gaussian data at a prime. For $d = 1$, $\mathbb{Z}[i]$ is a PID with unit group μ_4. For each odd prime $r \equiv 1 \pmod 4$, fix a Gaussian prime γ_r of norm r, so $r = \gamma_r \overline{\gamma_r}$. Unique factorization shows that every Gaussian integer of norm r^2 is a unit multiple of r, γ_r^2, or $\overline{\gamma_r}^2$. The first orbit gives the four axial representations of r^2 as a sum of two squares, while the other two give the eight non-axial ordered signed representations. Since γ_r^2 is not axial, $2\gamma_r^2$ has even nonzero coordinates; taking absolute values and ordering them gives integers $g_r, h_r > 0$ with

$$\begin{gathered} g_r^2 + h_r^2 = 4r^2, \quad 0 < g_r < h_r, \quad g_r, h_r \equiv 0 \pmod 2, \\ \xi_r = \arcsin(g_r/(2r)) \in (0, H/2). \end{gathered} \tag{16}$$

(Equality $g_r = h_r$ would force $g_r^2 = 2r^2$, which is impossible.) The two complementary chord half-angles at radius r are ξ_r and $H - \xi_r$. Let $w_{\xi_r} = e^{i(H-\xi_r)} = (g_r + ih_r)/(2r)$ be the number attached to the chord of half-angle ξ_r. Here $\kappa = 2$, $M = r$, and Lemma 4.1 gives $|v_r(w_{\xi_r})| = 1$; fix the order of the two prime ideals over r so that $v_r(w_{\xi_r}) = +1$.
Angle representation. Let w be the number of any non-diameter Gaussian chord at radius n, and let a_r be its components at the oriented split primes. The quotient $w / \prod_r w_{\xi_r}^{a_r}$ has trivial fractional ideal because all of its split-prime components vanish; hence it is a unit of $\mathbb{Z}[i]$, i.e. a power of i. Comparing arguments gives

$$\alpha = cH + \sum_r a_r \xi_r, \quad c \in \mathbb{Z}, \quad 0 < \alpha < H, \tag{17}$$

where a_r are the split-prime components with these orientations. The vector (a_r) is the component vector of the chord, so it is additive. A chord with all $a_r = 0$ would have β equal to the argument of a unit, which is impossible for $0 < \beta < H$.

Two distinct split primes. Let $n = pq$ with distinct primes $p, q \equiv 1 \pmod 4$ and write $x > y$ for the two angles ξ_p, ξ_q in decreasing order. (Since the radius is symmetric in its prime factors, we label the primes by angle size rather than numerical size.) They cannot be equal, by (14). By Lemma 4.1, $(a, b) \in \{-1, 0, 1\}^2$. For each of the eight nonzero pairs there is exactly one admissible c in (17), and the complete candidate list, with the weight $\Phi = 4c + 2a + b$, is

(a,b)	c	α	Φ	(a,b)	c	α	Φ
$(1,0)$	0	x	2	$(1,1)$	0	$x+y$	3
$(-1,0)$	1	$H-x$	2	$(-1,-1)$	1	$H-x-y$	1
$(0,1)$	0	y	1	$(1,-1)$	0	$x-y$	1
$(0,-1)$	1	$H-y$	3	$(-1,1)$	1	$H-x+y$	3

(Here $0 < y < x < H/2$, so $0 < x - y < H$ and $0 < x + y < H$.) Thus $\Phi \geq 1$ on every candidate and $\Phi \leq 3$. For a rational block, (14) gives $\sum a = \sum b = 0$, so the total is $T = H \sum c$ and $\sum \Phi = 4 \sum c = 4T/H = \frac{4}{3} T/U$. Hence a block of total tH has at most $4t$ sides, and a π-block at most eight. Equality is attained by doubling the H-block

$$(x - y) + y + y + (H - x - y) = H. \tag{18}$$

To verify actual integer chord lengths, label the primes so that $x = \xi_p > y = \xi_q$ (this labeling need not have $p < q$). With $n = pq$, the three distinct half-angles in (18) correspond respectively to the chord lengths

$$\frac{|g_p h_q - h_p g_q|}{2}, \qquad p g_q, \qquad \frac{h_p h_q - g_p g_q}{2}. \tag{19}$$

(For instance $2n \sin y = pq\, g_q/q = p g_q$ and $2n \sin(H - x - y) = 2n \cos(x + y) = (h_p h_q - g_p g_q)/2$.) The final numerator is positive because $x + y < H$. All three are integers by (16) (the numerators are divisible by 4), and their multiplicities in the octagon are $2, 4, 2$. Thus the eight-side upper bound is attained whenever $p \neq q$ both split in $\mathbb{Z}[i]$.

A repeated split prime. For $n = p^2$ with $p \equiv 1 \pmod 4$, put $x = \xi_p$. By Lemma 4.1, $a \in \{-2, -1, 1, 2\}$, and (17) gives exactly four candidates:

a	1	-1	2	-2
c	0	1	0	1
α	x	$H-x$	$2x$	$H-2x$
$\Phi = 3c + a$	1	2	2	1

(here $0 < 2x < H$). So $\Phi \in \{1, 2\}$, $\sum \Phi = 3 \sum c = 3T/H = T/U$, and a π-block has at most six sides.

Exactly one split prime. Suppose that exactly one odd prime r splits in $\mathbb{Z}[i]$ and that it divides n to exponent one, as for $n = 2r$ or $n = rq$ with $q \equiv 3 \pmod 4$ (this includes the one-split-prime case of odd semiprimes). The only nonzero component is $a = \pm 1$. Putting $x = \xi_r$, the complete candidate set is $\{x, H - x\}$, i.e. $(c, a) = (0, 1)$ and $(1, -1)$, and both candidates have value 1 under the weight $\Phi = 2c + a$. Hence a rational block with total T has exactly $\sum \Phi = 2T/H$ members, that is, at most $\frac{2}{3} T/U \leq T/U$. If no odd prime splits in $\mathbb{Z}[i]$ there is no non-diameter Gaussian chord at all.

4.4. **Eisenstein classes and integral constructions.** For two distinct split primes the list of candidate angles depends on whether $x+y<U$ or $x+y>U$ (notation below); only the first case allows eighteen sides.

Let $K=\mathbb{Q}(\sqrt{-3})$, $\mathcal{O}_K^\times=\mu_6$, $H=3U$. Its ring of integers is a PID. Every $\nu\in\mathcal{O}_K$ has the form $\nu=(s+u\sqrt{-3})/2$ with integers $s\equiv u \pmod 2$; conversely, any integer solution of $s^2+3u^2=4r^2$ satisfies $s\equiv u \pmod 2$ (reduce modulo 4), so it defines such a ν with $\mathrm{N}(\nu)=r^2$. A solution with $s,u>0$ corresponds to $0<\arg\nu<H$, and then $\tan\arg\nu=u\sqrt{3}/s$ is the $\tan\beta$ of (10).

Lemma 4.4 (Eisenstein chords at a prime radius)**.** *Let $r\equiv 1 \pmod 3$ be prime. The positive solutions of $s^2+3u^2=4r^2$ are $(s,u)=(r,r)$, with half-angle U, and exactly three further solutions, whose half-angles are*

$$x_r,\qquad 2U-x_r,\qquad 2U+x_r\qquad \text{for a unique } x_r\in(0,U).$$

In particular there is exactly one solution with $0<s<r$, namely the one with half-angle $x_r=\arcsin(s_r/(2r))$, and it has the smallest s among all nontrivial solutions. This proves the existence and uniqueness claimed before Theorem A.

Proof. Because $\mathcal{O}_K$ is a unique-factorization domain and $r=\varpi\bar\varpi$ for a prime ϖ, every ν of norm r^2 is associated to exactly one of r, ϖ^2, or $\bar\varpi^2$. Unit multiples of r give only the half-angle U in the open sector (namely $\arg\nu=2U$, i.e. $(s,u)=(r,r)$).

For the two nontrivial orbits, let $\phi\in(0,2U)$ be $2\arg\varpi$ reduced modulo $2U$. Neither $\phi=0$ nor $\phi=U$ is possible: either would make $\varpi^2/r=\varpi/\bar\varpi$ a root of unity, contradicting its nontrivial fractional ideal $(\varpi)/(\bar\varpi)$. Thus $\delta=\min\{\phi,2U-\phi\}$ belongs to $(0,U)$. Applying the six units, conjugating, and retaining arguments in $(0,H)=(0,3U)$ gives exactly three arguments, $\beta=\delta,\ 2U-\delta,\ 2U+\delta$. Their chord half-angles $\alpha=H-\beta$ are, in increasing order, $U-\delta,\ U+\delta,\ 3U-\delta$. Putting $x_r=U-\delta$ gives precisely $\{x_r,2U-x_r,2U+x_r\}$, where $0<x_r<U$. Thus exactly one nontrivial chord half-angle is below U; since $s=2r\sin\alpha$, it is the unique solution with $0<s<r$ and it has the smallest s among the nontrivial solutions. □

The choice of ϖ versus $\bar\varpi$ merely interchanges the nontrivial orbits. For a prime $r\equiv 1 \pmod 3$ we write (s_r,u_r) for the solution of Lemma 4.4 with half-angle x_r; thus

$$\begin{aligned} s_r^2+3u_r^2=4r^2,\quad s_r,u_r>0,\quad 0<s_r<r,\\ s_r\equiv u_r \pmod 2,\quad x_r=\arcsin(s_r/(2r)).\end{aligned}\tag{20}$$

Let $w_{x_r}=e^{i(H-x_r)}$ be the number of the chord of half-angle x_r at radius r. Then $r\nmid\gcd(s_r,u_r)$ (otherwise $(s_r/r,u_r/r)$ would be a positive solution of $a^2+3b^2=4$, i.e. the trivial solution $(1,1)$), so $r\mid M$ and $|v_r(w_{x_r})|=\operatorname{ord}_r M=1$ by the proof of Lemma 4.1. We orient the prime-ideal pair over r so that $v_r(w_{x_r})=+1$.

Angle representation. Let $p\neq q$ be distinct primes $\equiv 1 \pmod 3$ and write $x>y$ for x_p,x_q in decreasing order (their independence follows from (14)). For an arbitrary Eisenstein chord w at radius pq, write its oriented split-prime component vector $(a,b)\in\{-1,0,1\}^2$ and let $w_x=e^{i(H-x)}$, $w_y=e^{i(H-y)}$.

By the component identity (14), the quotient $w/(w_x^a w_y^b)$ is a unit, hence a sixth root of unity. Therefore for some $j \in \mathbb{Z}$ its complementary angle satisfies $\beta \equiv a(H-x)+b(H-y)+2jU \pmod{2\pi}$. Subtracting from $H = 3U$ and absorbing multiples of $2\pi = 12U$ into j gives $\alpha = cU + ax + by$ with $c = 3 - 3a - 3b - 2j$; so $c \equiv 1 - a - b \pmod 2$. This proves that every possible non-diameter Eisenstein half-angle lies in the candidate set

$$\begin{aligned} &\alpha = cU + ax + by, \quad (a,b) \in \{-1,0,1\}^2, \\ &c \in \mathbb{Z}, \quad c \equiv 1 - a - b \pmod 2, \quad 0 < \alpha < 3U. \end{aligned} \tag{21}$$

The finite list may contain unrealizable candidates, which is harmless for an upper bound.

Complete two-prime Eisenstein catalogue. Here $0 < y < x < U$, so $0 < x-y < U$ and $0 < x+y < 2U$. The equality $x+y = U$ is impossible: the half-angles x_p, x_q and the half-angle U of a hexagon side (whose component vector is zero) would satisfy $x_p + x_q - U = 0 \in \pi\mathbb{Q}$ with component vector $(1,1) \neq 0$, contradicting (14). Imposing the parity rule and $0 < cU + ax + by < 3U$ for each of the nine pairs (a,b) gives the table below, which lists the admissible values of c in each of the two chambers $x + y < U$ and $x + y > U$. In square brackets is the weight of the candidate: $\Phi_+ = 3c + a + b$ in the first chamber and $\Phi_- = c + a + b$ in the second.

(a,b)	chamber $x+y<U$ entries $c\,[\Phi_+]$	chamber $x+y>U$ entries $c\,[\Phi_-]$
$(-1,-1)$	$1\,[1],\ 3\,[7]$	$3\,[1]$
$(-1,1)$	$1\,[3],\ 3\,[9]$	$1\,[1],\ 3\,[3]$
$(0,0)$	$1\,[3]$	$1\,[1]$
$(1,-1)$	$1\,[3]$	$1\,[1]$
$(1,1)$	$1\,[5]$	$-1\,[1],\ 1\,[3]$
$(-1,0)$	$2\,[5]$	$2\,[1]$
$(0,-1)$	$2\,[5]$	$2\,[1]$
$(0,1)$	$0\,[1],\ 2\,[7]$	$0\,[1],\ 2\,[3]$
$(1,0)$	$0\,[1],\ 2\,[7]$	$0\,[1],\ 2\,[3]$

(Here $(0,0)$ is the hexagonal angle U, and $(a,b) = (0,1), (1,0)$ with $c = 0$ are y and x.) Each chamber has thirteen candidates. The weights in the first chamber are $1,1,1,3,3,3,5,5,5,7,7,7,9$ and those in the second are $1^{(9)}, 3^{(4)}$ (nine ones and four threes). In particular $\Phi_+ \geq 1$ with maximum 9, and $\Phi_- \geq 1$ with maximum 3. For a rational block, $\sum a = \sum b = 0$, so the total is $T = U \sum c$ and

$$\sum \Phi_+ = 3 \sum c = 3\,T/U, \qquad \sum \Phi_- = \sum c = T/U.$$

Hence $k \leq 3T/U$ if $x + y < U$ and $k \leq T/U$ if $x + y > U$.

The eighteen-gon. In the active case $x = x_p$, $y = x_q$ (in either order) and $x + y < U$, six copies of

$$x + y + (U - x - y) = U \tag{22}$$

form an 18-gon of radius $n = pq$. The first two chord lengths are qs_p and ps_q, and the third is

$$S = \frac{3u_pu_q - s_ps_q - 3(s_pu_q + u_ps_q)}{4}. \tag{23}$$

Indeed $2pq\sin x_p = qs_p$, $2pq\sin x_q = ps_q$, and, using $\sin x_r = s_r/(2r)$, $\cos x_r = \sqrt{3}u_r/(2r)$,

$$\begin{aligned}\sin(U - x_p - x_q) &= \tfrac{1}{2}\cos(x_p + x_q) - \tfrac{\sqrt{3}}{2}\sin(x_p + x_q)\\ &= \frac{3u_pu_q - s_ps_q - 3(s_pu_q + u_ps_q)}{8pq}.\end{aligned}$$

So $S = 2pq\sin(U - x_p - x_q)$, and it is positive because $U - x_p - x_q > 0$. Set $R = 4S$. Modulo 4, using $3 \equiv -1 \pmod 4$, one has

$$R \equiv (s_p - u_p)(u_q - s_q) \equiv 0 \pmod 4,$$

since $s_r \equiv u_r \pmod 2$. Thus $S \in \mathbb{Z}$, and (22) gives exact closure with six copies of each chord length.

One split prime, no split prime, and prime squares. If only one odd prime splits in $\mathbb{Q}(\sqrt{-3})$ (with exponent one in n), write $x = x_r$; the same argument as above gives $a \in \{-1, 0, 1\}$, $c \equiv 1 - a \pmod 2$ in (21), and the complete candidate set is $\{U,\ x,\ 2U-x,\ 2U+x\}$, i.e. $(c, a) = (1, 0), (0, 1), (2, -1), (2, 1)$ with weights $\Phi = c + a = 1, 1, 1, 3$. A rational block therefore has $k \le \sum \Phi = T/U$. If no odd prime splits, the only candidate is $\alpha = U$ (the hexagonal angle), again with $k \le T/U$.

For a repeated split prime $n = p^2$ with $x = x_p$, the coefficient a lies in $\{-2, -1, 0, 1, 2\}$. If $2x < U$, its complete candidate list is

$$(c, a) \in \{(1, -2), (3, -2), (2, -1), (1, 0), (0, 1), (2, 1), (1, 2)\}; \tag{24}$$

the weights $\Phi_+ = 3c + a$, in the displayed order, are $(1, 7, 5, 3, 1, 7, 5)$, all at least 1. If $2x > U$, the candidates are

$$(c, a) \in \{(3, -2), (2, -1), (1, 0), (0, 1), (2, 1), (-1, 2), (1, 2)\}; \tag{25}$$

the weights $\Phi_- = c + a$, in the displayed order, are $(1, 1, 1, 1, 3, 1, 3)$, all at least 1. Therefore the respective π-block bounds are 18 and 6 (because $\sum \Phi_+ = 3T/U$ and $\sum \Phi_- = T/U$), with maximal weights 7 and 3. The boundary $2x = U$ is impossible, since $\sin^2 x = s_p^2/(4p^2)$ is rational while $\sin^2(\pi/12) = (2-\sqrt{3})/4$ is not. When $2x < U$, six copies of $x+x+(U-2x) = U$ form an 18-gon: the first two equal chord lengths are ps_p, while the third is

$$S = \frac{2p^2 - s_p^2 - 3s_pu_p}{2}. \tag{26}$$

This is a positive integer because $s_p \equiv u_p \pmod 2$; thus the side multiset is $\{(ps_p)^{[12]}, S^{[6]}\}$.

4.5. **Mixtures, major arcs and diameters.** The catalogues are summarized in Table 1. The two last columns are the only information about them that is needed below: the *density* $\rho = \sum \Phi/(T/U)$ controls minor-arc polygons, and the maximal weight controls major-arc relations.

Class and case (n semiprime)	weight Φ	density ρ	$\max \Phi$
Gaussian, two distinct split primes	$4c + 2a + b$	4/3	3
Gaussian, $n = p^2$, $p \equiv 1$ (4)	$3c + a$	1	2
Gaussian, one split prime	$2c + a$	2/3	1
Eisenstein, two distinct split primes, $x + y < U$	$3c + a + b$	3	9
Eisenstein, two distinct split primes, $x + y > U$	$c + a + b$	1	3
Eisenstein, $n = p^2$, $p \equiv 1$ (3), $2x < U$	$3c + a$	3	7
Eisenstein, $n = p^2$, $p \equiv 1$ (3), $2x > U$	$c + a$	1	3
Eisenstein, one split prime	$c + a$	1	3
Eisenstein, no split prime	c	1	1
Ordinary ($d \neq 1, 3$)	(Lemma 4.3)	1	—

TABLE 1. Weights on the candidate half-angles. For a rational block with total T, $k \leq \sum \Phi = \rho T/U$, since every candidate has $\Phi \geq 1$ and $\sum a = \sum b = 0$. "Two distinct split primes" means $n = pq$ with $p \neq q$ both split in the given field; "one split prime" means that exactly one prime divisor of n splits, to exponent one.

Proposition 4.5 (Uniform semiprime density bounds)**.** *Let n be semiprime. Every rational block of k half-angles, pure in one squarefree class d and with total T, satisfies $k \leq \rho_d T/U$, where $U = \pi/6$ and*

$$\rho_d \leq \begin{cases} 3, & d = 3, \text{ two distinct split odd primes with } x + y < U, \\ & \quad \text{or one repeated split odd prime with } 2x < U, \\ 4/3, & d = 1, \text{ two distinct split odd primes}, \\ 1, & \text{otherwise.} \end{cases} \tag{27}$$

Consequently, any minor-arc polygon without diameter at radius n has at most $6 \max_d \rho_d$ sides.

Proof. For ordinary classes the bound is Lemma 4.3. For the Gaussian and Eisenstein classes it is the column ρ of Table 1, which was derived in §4.3 and §4.4 (the one-split-prime Gaussian density 2/3 is bounded by 1). Let P be a minor-arc polygon without diameter; its sides form a π-block. By Lemma 3.2, the restriction of this π-block to each nonempty squarefree class is a rational block, with total $T_d > 0$ and k_d members, say; then $k_d \leq \rho_d T_d/U$ and $\sum_d T_d = \pi$. Summing over the classes gives $k \leq (\max_d \rho_d)\pi/U = 6 \max_d \rho_d$. □

Lemma 4.6 (No mixed non-diameter major arcs)**.** *In a non-diameter major-arc equation $\alpha_* = \sum_{j=1}^{k-1} \alpha_j < H$, all sources and the target belong to the same squarefree class.*

Proof. Let d_* be the target class, and suppose the foreign sources (those not in d_*) have total angle E with $0 < E < H$ (each source is positive and $\sum \alpha_j = \alpha_* < H$). Apply Lemma 3.3 to $\alpha_* - \sum_j \alpha_j = 0$ with signs $+1$ for the

target and -1 for the sources. If $d_* \neq 3$, the signed subtotal in d_* lies in $H\mathbb{Z}$ (parts (i), (ii) of the lemma); but it equals $\alpha_* - \sum_{j \in d_*} \alpha_j = E \in (0, H)$, a contradiction. If $d_* = 3$, every foreign class $e \neq 3$ with sources has subtotal in $H\mathbb{Z}$ and positive, hence at least H; so $E \geq H$, also impossible. □

Lemma 4.7 (Uniform major-arc bounds)**.** *A non-diameter major-arc polygon at a semiprime radius has at most four sides in an ordinary or Gaussian class, at most ten in the active two-prime Eisenstein case, at most eight in the active repeated-prime Eisenstein case, and at most four in any other Eisenstein case.*

Proof. By Lemma 4.6 all sides lie in one class. Let k be the number of sides, so there are $k - 1 \geq 2$ sources.

Ordinary class. The source and target component vectors are nonzero: a zero component would make the complementary angle the argument of a unit, impossible for $0 < \beta < H$. In the augmented signed zero sum $\{-v_*, v_1, \ldots, v_{k-1}\}$, no proper nonempty source-only submultiset can have sum zero, since Lemma 4.3 would give it an angle sum at least H, whereas $\sum \alpha_j = \alpha_* < H$. A proper zero-sum submultiset containing $-v_*$ would leave such a source-only submultiset as its nonempty complement. Hence the augmented multiset is a minimal zero sum, and Lemma 4.2 gives $k \leq 4$ (and $k \leq 3$ for repeated-prime or $n = 4$ cases).

Gaussian and Eisenstein classes. The component equation is $a_* = \sum a_j$, $b_* = \sum b_j$ (where present), by (14). The genuine major-arc equation is $c_*U + a_*x + b_*y = \sum_j (c_jU + a_jx + b_jy)$, so, after subtracting the component identities, $(c_* - \sum_j c_j)U = 0$ and hence $c_* = \sum_j c_j$ (with H replacing U in the Gaussian case). Thus each weight Φ of Table 1 is *exactly additive*: $\Phi(\alpha_*) = \sum_j \Phi(\alpha_j)$. Every source contributes at least 1, so $k - 1 \leq \Phi(\alpha_*) \leq \max \Phi$. By Table 1, $\max \Phi = 3$ in the two-prime Gaussian case (so $k \leq 4$), 2 in the repeated-prime Gaussian case ($k \leq 3$), and 1 in the one-prime Gaussian case (so there is no major-arc polygon with at least two sources). The Eisenstein maxima are 9 (two-prime active), 7 (repeated active), and 3 in all other cases; hence $k \leq 10, 8, 4$, respectively. □

Lemma 4.8 (Polygons with a diameter)**.** *Let n be semiprime and $\rho_{\max} = \max_d \rho_d$ over the classes present. A polygon at radius n with a diameter has at most $1 + 3\rho_{\max}$ sides, that is, at most 10, 5, or 4 sides when $\rho_{\max} = 3$, 4/3, or 1, respectively. These numbers are strictly below the minor-arc bounds 18, 8, 6.*

Proof. The other sides form an H-block (Remark 2.2). Exactly as in the proof of Proposition 4.5, with total $H = 3U$ in place of $\pi = 6U$, it has at most $3\rho_{\max}$ members, i.e. at most 9, 4, or 3. □

4.6. **Assembly: proof of Theorem A.**
Small radii $2p$ and 4. For $n = 2p$, p odd, all classes satisfy the baseline density $\rho \leq 1$: ordinary classes by Lemma 4.3, and the Gaussian and Eisenstein classes by the one-split-prime and no-split-prime rows of Table 1 (the prime 2 neither splits in $\mathbb{Q}(i)$ nor in $\mathbb{Q}(\sqrt{-3})$). Hence minor-arc polygons have at most 6 sides; by Lemmas 4.7 and 4.8, non-diameter major-arc polygons and polygons with a diameter have at most four sides. Therefore $a(2p) \leq 6$, and

the regular hexagon attains six. For $n = 4$, the squarefree classes of the non-diameter lengths $s = 1, \dots, 7$ are, in order, $7, 15, 55, 3, 39, 7, 15$ (from $4n^2 - s^2 = 64 - s^2 = 63, 60, 55, 48, 39, 28, 15$). None is Gaussian, and the only Eisenstein half-angle is U (at $s = 4$). Every ordinary class has only the single possible split component at 2, with $|v_2| \le 2$, so Lemma 4.3 applies and gives density 1; ordinary major-arc polygons have at most three sides (Lemma 4.7), while no major-arc relation can arise from angles all equal to U (it would read $U = rU$ with $r \ge 2$). Thus again $a(4) = 6$.

Odd semiprimes. Let $n = pq$ with $p \le q$ odd primes. By Proposition 4.5 and Table 1, the largest class density $\rho_{\max}$ is

- 3, if $p \equiv q \equiv 1 \pmod 3$ and $x_p + x_q < U$ (for $p = q$ this reads $2x_p < U$);
- 4/3, if this fails but $p < q$ and $p \equiv q \equiv 1 \pmod 4$;
- 1, otherwise.

Indeed, the Eisenstein class has density 3 only in the two "active" cases of Table 1; the Gaussian density exceeds 1 only for two distinct split primes; and the Eisenstein case takes precedence when both conditions hold, because its density is larger. The equalities $x_p + x_q = U$ and $2x_p = U$ cannot occur (see §4.4), so the first alternative fails exactly when $x_p + x_q > U$, or when one of p, q is not $\equiv 1 \pmod 3$.

Proof of Theorem A. Upper bounds. By Proposition 4.5, a minor-arc polygon without diameter has at most $6\rho_{\max}$ sides, that is, at most 18, 8 or 6 sides according to the three alternatives above. For $n = 2p$ and $n = 4$ the bound is 6 by the first paragraph of this subsection. By Lemmas 2.1, 4.6, 4.7 and 4.8, neither major-arc polygons nor polygons with a diameter exceed these maxima.

Lower bounds. In the active Eisenstein cases the integral constructions (22) and (26) give 18 sides; otherwise two distinct Gaussian split primes give the integral 8-side construction (18). The regular hexagon yields six in all remaining cases, including $n = 4$ and $n = 2p$.

The integer criterion. By the computation after (23),

$$\sin(U - x_p - x_q) = \frac{3u_pu_q - s_ps_q - 3(s_pu_q + u_ps_q)}{8pq}.$$

Since $0 < x_p + x_q < 2U$, this sine is positive precisely when $x_p + x_q < U$; this is equivalent to the claimed integer criterion (4). The repeated-prime case is obtained by putting $p = q$.

The special values. If $p = 2$ or $p = 3$, then $p \not\equiv 1 \pmod 3$ and $p \not\equiv 1 \pmod 4$, so neither exceptional case occurs; hence $a(2q) = a(3q) = 6$ for every prime q, since the exceptional cases require *both* primes to be $\equiv 1 \pmod 3$ or both $\equiv 1 \pmod 4$ (and $a(pq)$ is symmetric in p, q). □

Remark 4.9 (Primes and $n = 1$). For $n = 1$ the only chords are $s = 1$ ($\alpha = U$) and the diameter $s = 2$, so $a(1) = 6$. For a prime $n = p$, Lemma 4.1(v) gives 1-dimensional component vectors with entries in $\{-1, 0, 1\}$, whose minimal zero sums have two members only (Lemma 4.2); so no ordinary class contains a rational block, and (Gaussian one-split-prime and Eisenstein rows of Table 1) every class has density at most 1. The arguments above then give $a(p) = 6$ (minor-arc polygons have at most 6 sides; a diameter plus an H-block has at most 4; major arcs at most 4).

5. Growth along fixed powers

Increasing the exponents at fixed prime support behaves differently from the semiprime case: the following construction gives unbounded side counts along N^e for every fixed $N \geq 2$. Write

$$4N^2 - 1 = dv^2, \tag{28}$$

where d is squarefree and $v \geq 1$. Put

$$z = 1 + v\sqrt{-d}, \qquad \mathrm{N}(z) = 4N^2,$$

and write

$$z^j = x_j + y_j\sqrt{-d} \qquad (j \geq 1).$$

Lemma 5.1 (The quadratic chord tower)**.** *For every $j \geq 1$, both x_j and y_j are divisible by 2^{j-1}. If*

$$X_j = \frac{x_j}{2^{j-1}}, \qquad Y_j = \frac{y_j}{2^{j-1}},$$

then

$$X_j^2 + dY_j^2 = 4N^{2j}. \tag{29}$$

In particular, $|X_j|$ is an integer chord length at radius N^j. Moreover, if

$$\theta = \arg z \in (0, \pi/2), \qquad \tau = \frac{2\theta}{\pi},$$

then τ is irrational and the associated half-angle is

$$\lambda_j := \alpha_{N^j}(|X_j|) = \arcsin|\cos(j\theta)|. \tag{30}$$

Proof. The real parts satisfy the trace recursion

$$x_{j+2} = 2x_{j+1} - 4N^2 x_j, \qquad x_0 = x_1 = 1.$$

With $X_j = x_j/2^{j-1}$ this becomes

$$X_{j+2} = X_{j+1} - N^2 X_j, \qquad X_1 = 1, \quad X_2 = 1 - 2N^2, \tag{31}$$

so all X_j are integers. The imaginary parts satisfy the same trace recursion, with $y_1 = v$ and $y_2 = 2v$; after division by 2^{j-1} the same integral recurrence applies. Hence $2^{j-1} \mid y_j$ (with $y_0 = 0$). Dividing $\mathrm{N}(z^j) = (4N^2)^j$ by 4^{j-1} proves (29).

Since $z/(2N) = e^{i\theta}$,

$$\frac{X_j}{2N^j} = \cos(j\theta),$$

which gives (30). If θ/π were rational, then $e^{i\theta}$ would be a root of unity and

$$2\cos\theta = \frac{1}{N}$$

would be a rational algebraic integer. It would therefore be an integer, which is impossible for $N \geq 2$. Hence θ/π and τ are irrational. Consequently $X_j \neq 0$, so this is a positive integer chord length. □

For $e \geq j$, scaling the chord in Lemma 5.1 gives the integer side

$$s_{e,j} = N^{e-j}|X_j| \tag{32}$$

at radius N^e, with the same half-angle λ_j.

We use one elementary fact about the Stern–Brocot path of an irrational number in $(0, 1)$.

Lemma 5.2 (Odd–odd bracketing edges)**.** *Let $\tau \in (0, 1)$ be irrational. Along its Stern–Brocot path there are infinitely many bracketing edges*

$$\frac{a_1}{b_1} < \tau < \frac{a_2}{b_2}, \qquad b_1a_2 - a_1b_2 = 1, \tag{33}$$

for which both numerators a_1 and a_2 are odd. The denominators of these edges are unbounded.

Proof. For neighbors $a_1/b_1 < a_2/b_2$ with $b_1a_2 - a_1b_2 = 1$ the numerators a_1, a_2 are not both even. Call such an edge *mixed* if exactly one numerator is odd. The mediant $(a_1 + a_2)/(b_1 + b_2)$ has odd numerator if the edge is mixed and even numerator if it is odd–odd. The Stern–Brocot path of τ starts at the mixed edge $0/1 < \tau < 1/1$. A step from a mixed edge replaces either the even endpoint, which gives an odd–odd edge, or the odd endpoint, which gives again a mixed edge with the same even endpoint ε. If the second alternative occurred at every step from some point on, the other endpoint would converge to ε and $\tau = \varepsilon$ would be rational. Hence every mixed edge is followed later by an odd–odd edge, and every odd–odd edge is followed immediately by a mixed edge. So odd–odd edges occur infinitely often. The edges are pairwise distinct and only finitely many fractions have bounded denominator, so the denominators are unbounded. □

Proof of Theorem B. Choose an odd–odd edge (33). Since $b_1a_2 - a_1b_2 = 1$,

$$0 < b_1\tau - a_1 < \frac{1}{b_2} \leq 1, \qquad 0 < a_2 - b_2\tau < \frac{1}{b_1} \leq 1.$$

Since a_1 and a_2 are odd and both distances are less than 1 (odd integers are 2 apart), these are the distances of $b_1\tau$ and $b_2\tau$ from the nearest odd integers a_1 and a_2. (If m is odd and $|t - m| \leq 1$, then $|\cos(Ht)| = |\sin(H(t - m))|$, and $\arcsin|\cos(Ht)| = H|t - m|$.) Formula (30) therefore gives, with $H = \pi/2$,

$$\lambda_{b_1} = H(b_1\tau - a_1), \qquad \lambda_{b_2} = H(a_2 - b_2\tau).$$

The determinant identity now yields

$$b_2\lambda_{b_1} + b_1\lambda_{b_2} = H(b_1a_2 - a_1b_2) = H. \tag{34}$$

Doubling (34) gives a π-block with $2(b_1 + b_2)$ sides. By (32), both required chord lengths are integers at the common radius N^e, where $e = \max\{b_1, b_2\}$. Hence

$$a(N^e) \geq 2(b_1 + b_2) \geq 2e. \tag{35}$$

By Lemma 5.2, $e = \max\{b_1, b_2\}$ is unbounded over the odd–odd edges, so (35) gives (5) and shows that $a(N^e)$ is unbounded. By Proposition 2.4, $a(N^e)$ is nondecreasing in e. Hence $a(N^e) \to \infty$. □

The following example makes the construction explicit; the proof above does not depend on it.

Example 5.3 (A thirty-eight-gon at radius 8192)**.** Take $N = 2$. Then $d = 15$, $v = 1$, and $z = 1 + \sqrt{-15}$. For

$$\tau = \frac{2 \arg z}{\pi}$$

one has the odd–odd Farey edge

$$\frac{5}{6} < \tau < \frac{11}{13}, \qquad 6 \cdot 11 - 5 \cdot 13 = 1.$$

The Farey inequalities can be certified without trigonometric approximation. Writing $\theta = \arg z = \arccos(1/4)$, we have $5/6 < \tau$ because $\cos(5\pi/12) = (\sqrt{6} - \sqrt{2})/4 > 1/4$; the final inequality follows by squaring from $48 < 49$. Likewise $\tau < 11/13$ since $\cos(11\pi/26) = \sin(\pi/13) < \pi/13 < 1/4$. For $N = 2$, the integer recurrence (31) begins

$$1, -7, -11, 17, 61, -7, -251, -223, 781, 1673, -1451, -8143, -2339,$$

so in particular the scaled power coefficients are exactly

$$\frac{\Re(z^6)}{2^5} = X_6 = -7, \qquad \frac{\Re(z^{13})}{2^{12}} = X_{13} = -2339.$$

At the common radius $2^{13} = 8192$, the first chord scales by 2^7 and has length 896, while the second has length 2339. Thus

$$26\alpha_{8192}(896) + 12\alpha_{8192}(2339) = \pi,$$

so

$$\{896^{[26]}, 2339^{[12]}\}$$

is a thirty-eight-gon and $a(8192) \geq 38$.

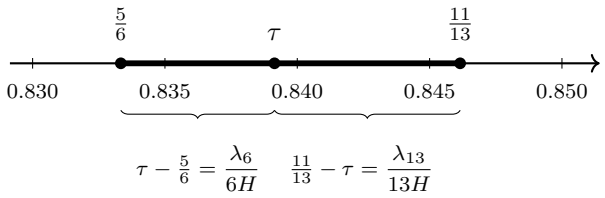


FIGURE 2. The Farey bracket $\frac{5}{6} < \tau < \frac{11}{13}$ of Example 5.3, drawn to scale; $\tau = 2\arg(1 + \sqrt{-15})/\pi \approx 0.83914$. Here $H = \pi/2$ and $\lambda_j = \alpha_{2^j}(|X_j|)$. The distances from τ to the endpoints are $\lambda_6/(6H)$ and $\lambda_{13}/(13H)$, and $6 \cdot 11 - 5 \cdot 13 = 1$ gives $13\lambda_6 + 6\lambda_{13} = H$. The plotted positions were rendered with AI-assisted TikZ code from these values.

6. Parity with one Gaussian split prime

We show that $a(n)$ is even if at most one prime divisor of n is congruent to 1 (mod 4). An odd extremal polygon is first reduced to a pure Gaussian block; the maximal length of such blocks is then computed by Farey neighbors.

Put $H = \pi/2$. A *Gaussian H-block* is an H-block (Definition 2.3) pure in the Gaussian class $d = 1$, and a *Gaussian π-block* is defined similarly. Let $H_G(n)$ and $P_G(n)$ be their maximum cardinalities at radius n, with value 0 if no such block exists (for instance if the class is absent).

Lemma 6.1 (Reduction of an odd maximum)**.** *If $a(n)$ is odd, then an extremal polygon is a pure Gaussian minor-arc polygon without diameter. In particular, $a(n) \le P_G(n)$.*

Proof. Let P be an extremal polygon with $k = a(n)$ sides, k odd. By Proposition 3.5, P must contain a non-diameter side of the Gaussian class. We treat the three kinds of polygons of Remark 2.2.

Polygon with a diameter. The remaining $k-1$ sides form an H-block with, say, $r = k-1 \ge 2$ members. Doubling that block gives a π-block, hence by Lemma 2.1 a polygon with $2r > r+1 = k$ sides, contradicting extremality.

Minor-arc polygon without diameter. The sides form a π-block. Apply Lemma 3.3 (with all $\varepsilon_i = 1$): the Gaussian subtotal $\mathcal{A}_1$ lies in $H\mathbb{Z}$. If other squarefree classes occur, the Gaussian subtotal and the non-Gaussian subtotal $\pi - \mathcal{A}_1 \in H\mathbb{Z}$ are both positive, hence both equal H. Let their side counts be k_G and k_N, so $k = k_G + k_N$. Doubling either H-block gives a π-block, hence an even polygon with $2k_G$ or $2k_N$ sides. Since k is odd, $k_G \ne k_N$, and

$$\max\{2k_G, 2k_N\} > k_G + k_N = k,$$

contradicting extremality. So no other class occurs: P is purely Gaussian.

Major-arc polygon. Here $\alpha_* = \sum_j \alpha_j < H$. Apply Lemma 3.3 to $\alpha_* - \sum_j \alpha_j = 0$: the Gaussian subtotal lies in $H\mathbb{Z}$. If the target is not Gaussian, this subtotal is $-\sum_{j\ \text{Gaussian}} \alpha_j \in (-H, 0]$, hence zero; thus there is no Gaussian side, contradicting the first paragraph of this proof. If the target is Gaussian, the Gaussian subtotal equals $\alpha_* - \sum_{j\ \text{Gaussian}} \alpha_j = \sum_{j\ \text{non-Gaussian}} \alpha_j \in [0, H)$, hence zero, so every source side is Gaussian and the relation is purely Gaussian. Write $4n^2 - s_*^2 = u_*^2$, where s_* is the side with half-angle α_*; the complementary chord u_* is an integer, is not a diameter, and has half-angle $H - \alpha_*$. The source sides together with this complementary chord form an H-block with $(k-1)+1 = k$ members; doubling it gives a polygon with $2k > k$ sides, a contradiction.

Hence P is a pure Gaussian π-block without diameter, and so $k \le P_G(n)$. □

Assume now that n has exactly one distinct prime divisor $q \equiv 1 \pmod 4$, and write $q^e \| n$. Choose a Gaussian prime ϖ with $\varpi\overline{\varpi} = q$ whose argument satisfies

$$\vartheta = \arg \varpi \in (0, \pi/4); \qquad \sigma = \frac{4\vartheta}{\pi} \in (0,1) \tag{36}$$

(of the eight numbers $u\varpi$, $u\overline{\varpi}$, $u \in \mu_4$, exactly one has argument in $(0, \pi/4)$). The number σ is irrational. Indeed, a rational value would make $\varpi/\overline{\varpi} = e^{2i\vartheta}$ a root of unity in $\mathbb{Q}(i)$, i.e. a power of i; since $0 < 2\vartheta < \pi/2$, this is impossible.

For $1 \le j \le e$, let

$$\ell_j = \lfloor j\sigma \rfloor, \qquad f_j = j\sigma - \ell_j \in (0,1).$$

Lemma 6.2 (The rank-one Gaussian angle set)**.** *The non-diameter Gaussian half-angles at radius n are exactly*

$$Hf_j, \qquad H(1-f_j), \qquad 1 \le j \le e. \tag{37}$$

Proof. Let $s+iu$ be a Gaussian integer with $s^2+u^2=(2n)^2$. A prime $r\equiv 3 \pmod 4$ is inert in $\mathbb{Z}[i]$, so its exponent in $s+iu$ is forced and contributes no argument. The prime 2 is ramified and contributes only a multiple of H, which can be absorbed into the unit. For the unique split prime $q=\varpi\overline{\varpi}$, write the exponents of ϖ and $\overline{\varpi}$ in $s+iu$ as a and $2e-a$. Their contribution to the argument is

$$(a-(2e-a))\vartheta = 2(a-e)\vartheta.$$

Thus, modulo H, the argument of $s+iu$ is $\pm 2j\vartheta = \pm Hj\sigma$ with $j=|a-e|\in\{0,\dots,e\}$; the case $j=0$ would give an axial vector, that is a diameter or a degenerate side, which is excluded. The half-angle of the chord is H minus this argument, and folding into $(0,H)$ gives Hf_j or $H(1-f_j)$ with $1\le j\le e$. Conversely, for $1\le j\le e$ the Gaussian integer $\varpi^{e+j}\overline{\varpi}^{\,e-j}$, multiplied by the forced factors at 2 and at the primes $\equiv 3 \pmod 4$ and by a unit, has norm $(2n)^2$ and positive real and imaginary parts (it is not axial, because σ is irrational), so it defines a chord; its half-angle is $H(1-f_j)$, and exchanging ϖ and $\overline{\varpi}$ gives Hf_j. None of these angles is 0 or H, since σ is irrational. □

The primes 2 and those congruent to 3 (mod 4) therefore change only the rational scale of the corresponding chords; they create no new Gaussian direction.

Let

$$\frac{a_1}{b_1} < \sigma < \frac{a_2}{b_2} \tag{38}$$

be the two neighbors of σ in the Farey sequence of order e. Then

$$b_1a_2 - a_1b_2 = 1, \qquad b_1,b_2\le e, \qquad b_1+b_2 > e.$$

Indeed, if $b_1+b_2\le e$, the mediant $(a_1+a_2)/(b_1+b_2)$ would be a fraction of the Farey sequence of order e strictly between two consecutive terms. Put

$$D=b_1+b_2, \qquad C=a_1+a_2.$$

Lemma 6.3 (Exact Gaussian block maxima)**.** *With the notation above,*

$$H_G(n)=D, \qquad P_G(n)=2D. \tag{39}$$

Proof. Represent the two angle types in (37) by

$$v_j=(-\ell_j,j), \qquad v_j'=(\ell_j+1,-j),$$

so that a vector $(\mu,\nu)\in\mathbb{Z}^2$ represents the angle $H(\mu+\nu\sigma)$. Because 1 and σ are linearly independent over $\mathbb{Q}$, an H-block sums to $(1,0)$ and a π-block to $(2,0)$.

The mediant C/D lies strictly between the Farey neighbors (38). For every $1\le j\le e$, the function $x\mapsto\lfloor jx\rfloor$ is constant on this Farey interval: a change would occur at a rational with denominator at most $j\le e$, contradicting that the two endpoints are adjacent in the Farey sequence of order e. Moreover, $\gcd(C,D)=1$ by the determinant-one identity and $D>e\ge j$, so Cj/D is not an integer. Consequently

$$\ell_j=\left\lfloor \frac{Cj}{D}\right\rfloor,$$

and hence

$$1\le Cj-D\ell_j\le D-1. \tag{40}$$

Define the integral linear functional

$$F(\mu,\nu) = D\mu + C\nu.$$

By (40), every available vector v_j and v'_j has $1 \le F \le D-1$ (indeed $F(v_j) = Cj - D\ell_j$ and $F(v'_j) = D - F(v_j)$). Since $F(1,0) = D$ and $F(2,0) = 2D$, an H-block has at most D sides and a π-block at most $2D$ sides.

For the lower bound, Farey adjacency gives

$$b_2(-a_1, b_1) + b_1(a_2, -b_2) = (1, 0).$$

Here $(-a_1, b_1) = v_{b_1}$ and $(a_2, -b_2) = v'_{b_2}$, because $a_1/b_1 < \sigma < a_2/b_2$ gives $\ell_{b_1} = a_1$ and $\ell_{b_2} = a_2 - 1$. Hence b_2 copies of the first angle and b_1 copies of the second form an H-block with $b_1 + b_2 = D$ sides. Doubling it gives a π-block with $2D$ sides. This proves (39). □

Proof of Theorem C. If $\omega_1(n) = 0$, Theorem 3.6 (or Proposition 3.5) shows that every admissible polygon already has even side count. Suppose $\omega_1(n) = 1$. If $a(n)$ were odd, Lemma 6.1 would give $a(n) \le P_G(n)$. But Lemma 6.3 shows that $P_G(n) = 2D \ge 4$ is attained by a Gaussian π-block, which is itself an admissible polygon, so $a(n) \ge P_G(n)$. Thus an odd maximum would have to equal the even number $P_G(n)$, a contradiction. Therefore $a(n)$ is even. □

Remark 6.4. The hypothesis $\omega_1(n) \le 1$ is used in Lemma 6.2, where the Gaussian angles are indexed by a single exponent. For several distinct primes $q \equiv 1 \pmod 4$ they are indexed by a multidimensional box, and the functional F would have to be replaced by a higher-dimensional zero-sum argument. We do not know whether $a(n)$ is even for every n.

7. Examples and consequences

Figure 3 shows the polygons of the following two examples, which realize the lower bounds of Theorem A.

Example 7.1 (Eisenstein jump)**.** For $p = 7$ and $q = 13$,

$$(s_7, u_7) = (2, 8), \qquad (s_{13}, u_{13}) = (1, 15).$$

The first two chord lengths are $qs_p = 26$ and $ps_q = 7$, and formula (23) gives $S = 61$. Thus

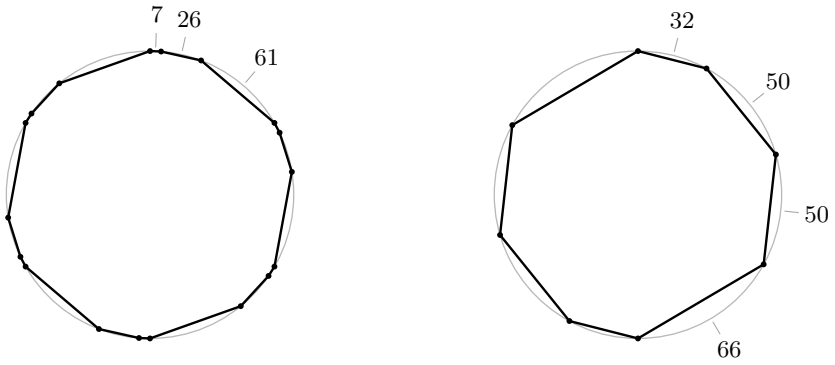
$$\{7^{[6]}, 26^{[6]}, 61^{[6]}\}$$

is the side multiset of an eighteen-gon of radius 91; hence $a(91) = 18$ by Theorem A.

Example 7.2 (Gaussian jump)**.** For $p = 5$ and $q = 13$, take $(g_5, h_5) = (6, 8)$ and $(g_{13}, h_{13}) = (10, 24)$ from (16); here $x = \xi_5 > y = \xi_{13}$. Formula (19) gives the integer side lengths $|6 \cdot 24 - 8 \cdot 10|/2 = 32$, $5 \cdot 10 = 50$, and $(8 \cdot 24 - 6 \cdot 10)/2 = 66$. Doubling the H-block (18) therefore gives the side multiset

$$\{32^{[2]}, 50^{[4]}, 66^{[2]}\},$$

so $a(65) = 8$ by Theorem A.



(a) Radius 91: sides $7^{[6]}, 26^{[6]}, 61^{[6]}$. (b) Radius 65: sides $32^{[2]}, 50^{[4]}, 66^{[2]}$.

FIGURE 3. The polygons of Examples 7.1 and 7.2, not drawn to the same scale. Vertices are placed according to the central angles $2\alpha_n(s)$ of the consecutive sides, which repeat the pattern $7, 26, 61$ (six times) in (a) and $32, 50, 50, 66$ (twice) in (b); one period is labelled. The diagrams were rendered with AI-assisted TikZ code from these chord lengths.

Corollary 7.3. *For a prime p,*

$$a(p^2) = 18$$

exactly when $p \equiv 1 \pmod 3$ and $2x_p < \pi/6$; otherwise $a(p^2) = 6$. In particular,

$$a(49) = 18, \qquad a(169) = 18, \qquad a(361) = 6.$$

Proof. The prime-square criterion follows from Theorem A. For the three stated examples, the positive solutions of $s^2 + 3u^2 = 4p^2$ at $p = 7, 13, 19$ are, respectively,

$$\begin{aligned} p = 7: &\quad (2,8), (7,7), (11,5), (13,3), \\ p = 13: &\quad (1,15), (13,13), (22,8), (23,7), \\ p = 19: &\quad (11,21), (19,19), (26,16), (37,5). \end{aligned}$$

Their completeness follows from Lemma 4.4 (one trivial solution (p,p) and three nontrivial ones). Hence the small-angle representatives are $(s_p, u_p) = (2,8), (1,15), (11,21)$. Since $3 < (7/4)^2$ and $3 > (12/7)^2$,

$$\frac{1}{16} < \sin^2(\pi/12) = \frac{2-\sqrt{3}}{4} < \frac{1}{14},$$

the rational inequalities $1/49, 1/676 < 1/16$ and $121/1444 > 1/14$ show exactly that $2x_7, 2x_{13} < \pi/6 < 2x_{19}$. □

By Theorems B and C, for every prime p the numbers $a(p^e)$ are even and tend to infinity with e.

8. Open questions

By Theorem C the number $a(n)$ is even when $\omega_1(n) \leq 1$, and by Theorem A it is even for every semiprime n. We do not know whether

$$a(n) \equiv 0 \pmod 2 \qquad (n \geq 1).$$

The OEIS entry [5] states that $a(n)$ is even for $n \leq 400$; this is not used in the proofs. If several distinct primes $\equiv 1 \pmod 4$ divide n, the Gaussian component vectors are no longer collinear and the Farey argument of Section 6 does not apply.

Theorem B gives $\limsup a(N^e)/e \geq 2$ but no upper bound. The order of growth of $a(N^e)$ is unknown for every $N \geq 2$, already for $a(2^e)$. It is also open how the factorization of n determines $a(n)$ beyond the semiprime case.

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